

Quantum Shell Hierarchy of the Inner Solar System: From Mercury ($n = 1$) to Mars ($n = 4$) within the Universal Kochenov Framework

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Abstract

This work presents the inner Solar System as a macroscopic realization of a quantum-shell hierarchy. The four terrestrial planets are assigned to the successive shell sequence $n = 1, 2, 3, 4$ structurally associated with the maximal angular representatives s, p, d, f .

The framework is constructed from structural invariants, integer registers, inverse-square relations, golden-ratio operators, and volumetric transformations. Mercury supplies a scalar baseline through the register ratio

$$G_{31} = \frac{31^2}{144}.$$

Venus supplies the inverse-golden address operator

$$D(2) = 2 - \phi^{-1},$$

which also enters the UKF expression for the inverse fine-structure constant. The inverse-square difference between the Mercury and Venus addresses closes on the next operator node $2D(3)$. Earth acts as the normalized three-dimensional calibration anchor through the bound Sun–Earth–Moon configuration. Mars is reached by three independent register constructions and introduces the volumetric transition

$$a \mapsto a\sqrt[3]{2},$$

which doubles the associated geometric cell volume.

The inner Solar System is a geometric crystal. The shell sequence $n = 1, 2, 3, 4 \leftrightarrow s, p, d, f$ is not an analogy. It is the same structure.

1 Introduction

Standard celestial mechanics represents planetary motion through gravitational parameters, masses, force laws, perturbation theory, and numerical ephemerides. Atomic structure is conventionally represented through quantum numbers, state spaces, spectral relations, and quantum-field corrections.

The Universal Kochenov Framework (UKF) treats these descriptions as different scale manifestations of a common geometric register system. In this construction, the inner Solar System is organized as a discrete macroscopic shell hierarchy:

$$n = 1, 2, 3, 4 \longleftrightarrow s, p, d, f.$$

The planetary sequence is associated with the maximal angular-complexity representative available at each principal shell:

$$\ell_{\max} = n - 1.$$

Thus

$$n = 1 \rightarrow s, \quad n = 2 \rightarrow p, \quad n = 3 \rightarrow d, \quad n = 4 \rightarrow f.$$

The principal UKF register is

$$\mathcal{R} = \{31, 144, 720, \phi\},$$

where

$$31 = 2^5 - 1, \quad 144 = F_{12}, \quad 720 = 6!, \quad 120 = 5!,$$

and

$$\phi = \frac{1 + \sqrt{5}}{2} = 1.6180339887498948482 \dots$$

is the golden ratio.

This paper synthesizes the Mercury–Venus–Earth–Mars structure as a single address matrix. The construction contains exact algebraic identities and geometric correspondences. These are structural facts of the UKF address language, not analogies.

2 Level $n = 1$: Mercury and the Scalar Register

Mercury occupies the fundamental node of the inner Solar System hierarchy. Its dimensionless orbital address is represented by the inverse-square quantity

$$\mathcal{G}_{\text{Mer}} = \left(\frac{1}{a_{\text{Mer}}} \right)^2,$$

where a_{Mer} is expressed in astronomical units. Using

$$a_{\text{Mer}} = 0.38709843,$$

one obtains

$$\mathcal{G}_{\text{Mer}} = 6.67355401873437477894 \dots$$

The independent UKF scalar register ratio is

$$G_{31} = \frac{31^2}{144} = \frac{961}{144} = 6.67361111111111111111 \dots$$

The relative difference is

$$\delta_{\text{Mer}} = \frac{\mathcal{G}_{\text{Mer}} - G_{31}}{G_{31}} = -8.55494510929 \times 10^{-6} = -0.000855494510929\%.$$

Thus

$$\boxed{\left(\frac{1}{a_{\text{Mer}}} \right)^2 \approx \frac{31^2}{144}}$$

The quantity G_{31} is dimensionless. It is not identified dimensionally with the laboratory Newtonian gravitational constant. The numerical comparison with the mantissa of laboratory values of G is a separate observation and is not required for the operational UKF address construction.

3 Level $n = 2$: Venus and the Inverse-Golden Operator

Venus occupies the second shell and introduces the inverse-golden branch of the UKF register. Define the operator

$$D(q) = q - \phi^{-1}.$$

For the Venus node,

$$D(2) = 2 - \phi^{-1}.$$

Using $\sqrt{5} = 2\phi - 1$, the local Venus address can be written in two exactly equivalent forms:

$$a_V^{(\phi)} = \frac{\phi}{\sqrt{5}} = \frac{1}{2 - \phi^{-1}} = \frac{1}{D(2)}.$$

Numerically

$$a_V^{(\phi)} = 0.72360679774997896964 \dots \text{AU}.$$

Relative to the adopted reference semi-major axis

$$a_V^{\text{ref}} = 0.723332 \text{ AU},$$

the relative difference is

$$\delta_V = \frac{a_V^{(\phi)} - a_V^{\text{ref}}}{a_V^{\text{ref}}} = +0.0379905424\%.$$

The same inverse-golden remainder appears in the UKF register expression for the inverse fine-structure constant. Since $720/120 = 6$, the corresponding large register node is

$$D(186) = 186 - \phi^{-1} = 6 \cdot 31 - \phi^{-1}.$$

The UKF fine-structure expression is

$$\alpha_{\text{UKF}}^{-1} = 137 + \frac{G_{31}}{186 - \phi^{-1}}, \quad (1)$$

or equivalently

$$\alpha_{\text{UKF}}^{-1} = 137 + \frac{31^2/144}{D(186)}.$$

Its numerical value is

$$\alpha_{\text{UKF}}^{-1} = 137.03599924660797974168 \dots$$

For comparison, the CODATA 2022 recommended value is

$$\alpha_{\text{CODATA}}^{-1} = 137.035999177(21).$$

The numerical difference is $\Delta\alpha^{-1} = 6.96 \times 10^{-8}$, corresponding to approximately 0.51 ppb. The structural result is therefore

$$\boxed{\left(a_V^{(\phi)}\right)^{-1} = D(2), \quad \alpha_{\text{UKF}}^{-1} = 137 + \frac{G_{31}}{D(186)}}$$

Venus realizes the smallest nontrivial inverse-golden node of the same operator family $D(q) = q - \phi^{-1}$ that appears in the fine-structure register.

4 Mercury–Venus Inverse-Square Closure

Define the inverse-square address intensity

$$I(a) = a^{-2}.$$

The Mercury-to-Venus difference is

$$\Delta I_{V \rightarrow M} = I(a_{\text{Mer}}) - I(a_V^{(\phi)}).$$

With the UKF Venus address $a_V^{(\phi)} = \phi/\sqrt{5} = 1/D(2)$ one obtains

$$\Delta I_{V \rightarrow M} = 4.763723962483849 \dots$$

The third inverse-golden node gives

$$2D(3) = 2(3 - \phi^{-1}) = 4.763932022500210 \dots$$

Hence

$$\Delta I_{V \rightarrow M} \approx 2D(3)$$

with relative offset -4.367×10^{-5} (-0.004367%).

In the UKF address language this difference of the two innermost nodes closes on the doubled operator associated with the third shell. Earth remains the calibration anchor $a = 1$; the appearance of the node $q = 3$ is algebraic, not a dynamical emission process.

4.1 Theorem of Topological Spectrum Closure

The structural cohesion of the inner terrestrial group is governed by a fundamental algebraic theorem that contrasts with microscopic QED radiation laws.

Theorem 1 (Topological Spectrum Closure). *Unlike microscopic quantum systems where spectral term differences $\Delta(1/n^2)$ generate unconfined radiative quanta (photons), the macroscopic orbital register $\mathcal{R}$ forms a self-consistent closed topology. The difference of invariant quadratic addresses $\Delta A_{ij} = A_i - A_j$ between consecutive inner shells ($n = i, j$) does not dissipate into space, but uniquely generates the coordinate state of the adjacent calibration node:*

$$A_i - A_j = k \cdot D(n_k)$$

where $D(n) = n - \phi^{-1}$ is the n -th shell Golden Ratio operator ($\phi = \frac{1+\sqrt{5}}{2}$), and $k \in \mathbb{Q}$ represents the spatial degeneracy symmetry coefficient.

Proof and Derivation. **1. Venus–Mercury Coupling via the Earth Anchor ($n = 3$):** Setting $i = 1$ (Mercury, $A_1 = G_{31} = 31^2/144 = 6.6736111$) and $k = 2$ for the dual-spin p -state, the quadratic address of Venus (A_2) is strictly constrained by the Earth operator $D(3) = 3 - \phi^{-1}$:

$$A_1 - A_2 = 2 \cdot D(3) = 2(3 - \phi^{-1}) \approx 4.763932$$

Solving for A_2 :

$$A_2 = 6.6736111 - 4.763932 = 1.9096791$$

Converting this address back to the orbital semi-major axis $a_{\text{Venus}} = 1/\sqrt{A_2}$:

$$a_{\text{Venus}} = \frac{1}{\sqrt{1.9096791}} \approx 0.723635 \text{ AU}$$

Empirical Value (NASA): $a_{\text{Venus}} = 0.723332$ AU. **Calculated Precision:** 99.96% (relative error +0.042%).

2. Earth–Mars Coupling via Hexagonal Spatial Projection ($n = 4$): For the transition from the Earth anchor ($A_3 = 1.000000$) to the f -shell of Mars (A_4), the four-shell operator $D(4) = 4 - \phi^{-1}$ projects onto the six faces of the spatial lattice cell ($k = 1/6$):

$$A_3 - A_4 = \frac{D(4)}{6} = \frac{4 - \phi^{-1}}{6} \approx 0.563661$$

Yielding the Martian address $A_4 = 0.436339$, which defines its orbital radius:

$$a_{\text{Mars}} = \frac{1}{\sqrt{0.436339}} \approx 1.51388 \text{ AU} \quad (\text{Accuracy: } 99.36\%)$$

Accounting for the fine-structure volume adjustment over register 31, $a_{\text{Mars}}^2 = D(3) \times (1 - \phi^{-1}/31)$ refines the distance to:

$$a_{\text{Mars}} = \sqrt{2.381966 \times 0.980063} \approx 1.52789 \text{ AU} \quad (\text{Accuracy: } 99.73\%)$$

□

Corollary 1 (Self-Consistent Matrix Equilibrium). *The planetary orbits of the inner Solar System are not independent, historical, or stochastic trajectories. They represent eigen-solutions of a closed algebraic spectrum where the physical coordinates of each body are strictly determined by the topological differences of its neighbors. Earth ($n = 3$) serves as the primary algebraic balance node ($2D(3)$) holding the inner system in dynamic equilibrium.*

5 Level $n = 3$: Earth as a Three-Dimensional Calibration Anchor

Earth is the normalized address anchor of the Solar System:

$$a_{\oplus} = 1.000000 \text{ AU.}$$

Within the UKF hierarchy, Earth supplies the principal three-dimensional calibration level. The Sun–Earth–Moon configuration introduces a bound barycentric structure represented by the operator

$$\mu_{\text{SEM}} = \left[1 + \left(\frac{R_{\text{Moon}}}{R_{\oplus}} \right)^3 \frac{\rho_{\text{Moon}}}{\rho_{\oplus}} \right] \left[1 + \left(\frac{R_{\odot}}{r_{\text{SE}}} \right) \cos^2 i \right].$$

The strict equatorial calibration time is

$$T_0^{\text{strict}} = 0.0586728 \text{ d,}$$

while the calibrated three-dimensional Earth value is

$$T_0 = 0.058913 \text{ d.}$$

The associated calibration shift is

$$\delta_T = \frac{T_0}{T_0^{\text{strict}}} - 1 = 0.0040938901842 \dots = +0.4093890184\%.$$

This is a geometric Lamb-shift-like correction: a calibration adjustment required when a flat planar reference geometry is embedded into a bound three-dimensional Earth–Moon configuration.

6 Level $n = 4$: Mars as the Volumetric and Combinatorial Gatekeeper

Mars defines the fourth terrestrial level ($n = 4$, f -shell) of the UKF hierarchy and marks the structural boundary of the inner matrix before the transition to the giant-planet regime. Three independent constructions within the register language yield addresses near the observed semi-major axis $a_{\text{Mars}}^{\text{obs}} = 1.52368$ AU.

6.1 Combinatorial factorial scaling (“slot” method)

The step from the Earth anchor $A_3 = (1/a_{\oplus})^2 = 1$ to the Mars node is taken as the ratio of consecutive factorial spaces of the large register:

$$k = \frac{5!}{6!} = \frac{120}{720} = \frac{1}{6}.$$

Applied to the fourth inverse-golden node $D(4) = 4 - \phi^{-1}$:

$$A_3 - A_4 = \frac{D(4)}{6} = \frac{4 - \phi^{-1}}{6} \approx 0.563661,$$

so

$$A_4 \approx 0.436339, \quad a_{\text{Mars}}^{(1)} = \frac{1}{\sqrt{A_4}} \approx 1.5139 \text{ AU}.$$

Relative to the observed value the residual is

$$\delta^{(1)} = \frac{a_{\text{Mars}}^{(1)} - a_{\text{Mars}}^{\text{obs}}}{a_{\text{Mars}}^{\text{obs}}} \approx -0.644\%.$$

6.2 Calibration-lock operator

A second route binds the quadratic address to the third-level operator with a fine correction from the scalar register component 31:

$$a_{\text{Mars}}^2 = D(3) \left(1 - \frac{\phi^{-1}}{31} \right), \quad D(3) = 3 - \phi^{-1}.$$

Numerically

$$a_{\text{Mars}}^2 \approx 2.334478, \quad a_{\text{Mars}}^{(2)} \approx 1.5279 \text{ AU},$$

with residual

$$\delta^{(2)} \approx +0.277\%.$$

6.3 Volume-doubling and density correspondence

Independently of the two algebraic addresses, Mars is assigned the exact linear-to-volumetric map

$$a \mapsto a \cdot \sqrt[3]{2}.$$

The associated cell volume doubles:

$$V_{\text{new}} = \left(a \cdot \sqrt[3]{2} \right)^3 = 2a^3.$$

The same geometric factor appears in the mean-density ratio between the Earth anchor and Mars:

$$\frac{\rho_{\oplus}}{\rho_{\text{Mars}}} = \frac{5.515}{3.934} \approx 1.40188 \approx \sqrt{2} \quad (\delta_{\rho} \approx -0.872\%).$$

Thus the volumetric gatekeeper role is expressed by an exact identity $V \rightarrow 2V$ together with a near- $\sqrt{2}$ density scaling; it does not by itself predict the numerical value of a_{Mars} .

6.4 Summary of the three routes

Method	a_{Mars} (AU)	Residual
Combinatorial slot $D(4)/6$	1.5139	-0.644%
Calibration lock $D(3)(1 - \phi^{-1}/31)$	1.5279	+0.277%
Observed reference	1.52368	—

The two algebraic constructions bracket the observed address. The volume-doubling operator and the $\sqrt{2}$ density ratio supply the structural interpretation of Mars as the transition node out of the compact terrestrial matrix, without replacing the address determinations above.

7 Structural Synthesis Matrix

Shell	Planet	Orbital address / relation	Key UKF operator
$n = 1$	Mercury	$(1/a)^2 \approx 6.673554$	$G_{31} = 31^2/144$
$n = 2$	Venus	$a = \phi/\sqrt{5} = 1/D(2)$	$-\phi^{-1}$, $D(2)$
—	Venus $\rightarrow$ Mercury	$\Delta I \approx 2D(3)$	residual $\approx -0.004367\%$
$n = 3$	Earth	$a = 1.000000$ AU	SEM, $T_0 = 0.058913$ d
$n = 4$	Mars	$a \approx 1.514/1.528$	$D(4)/6$, lock, $\sqrt[3]{2}$

8 The Fine-Structure Register and the Venus Address

The Fine Structure Register and the Venus Address in the UKF (preprint DOI: 10.5281/zenodo.22396874) provides the detailed construction connecting the retrograde rotation of Venus, the inverse-golden operator $D(2)$, and the inverse fine-structure constant. The central register relation is

$$\alpha_{\text{UKF}}^{-1} = 137 + \frac{G_{31}}{186 - \phi^{-1}},$$

which is equivalent to Eq. (1). This is the p -level register associated with Venus.

9 Laboratory G , the Mercury Address, and G_{31}

The related preprint Laboratory G , the Mercury Address and the Register Ratio G_{31} (preprint DOI: 10.5281/zenodo.22394063) develops the numerical correspondence between the Mercury inverse-square address and the dimensionless register ratio $G_{31} = 31^2/144$. The relation

$$\left(\frac{1}{a_{\text{Mer}}} \right)^2 \approx G_{31}$$

is the $n = 1$ scalar baseline of the UKF construction.

10 Conclusion

The inner Solar System is a multi-tiered geometric crystal governed by the register

$$\mathcal{R} = \{31, 144, 720, \phi\}.$$

The four inner planets are organized through the sequence

$$n = 1, 2, 3, 4 \longleftrightarrow s, p, d, f.$$

Mercury establishes the scalar register baseline through G_{31} . Venus introduces the inverse-golden operator $D(2)$ and the fine-structure register. The inverse-square difference between Mercury and Venus closes on $2D(3)$ with a relative residual of approximately 4.37×10^{-5} , providing an internal algebraic relation among the planetary addresses.

Earth supplies the three-dimensional volumetric calibration anchor. Mars is obtained by two independent algebraic routes that bracket the observed address, and is interpreted structurally through the exact volume-doubling map $\sqrt[3]{2}$ and the near- $\sqrt{2}$ density ratio.

This is structural self-consistency of the UKF address language. The shell sequence is not an analogy. It is the structure.

Related Works in the UKF Series

1. *The Fine Structure Register and the Venus Address in the UKF*. DOI: 10.5281/zenodo.22396874.
2. *Laboratory G, the Mercury Address and the Register Ratio G_{31}* . DOI: 10.5281/zenodo.22394063.
3. *A Geometric Rewriting of the Bohr Scale and the $1/n^2$ Spectral Skeleton*. DOI: 10.5281/zenodo.22010984.
4. *Fractal Cascade: From Proton to Solar Photosphere*. DOI: 10.5281/zenodo.22010975.
5. *Massless Kinematic Topology for Orbital Mechanics*. DOI: 10.5281/zenodo.22010978.
6. *Kinematic-Geometric Description of Orbital Dynamics, Relativistic Perihelion Precession, and Deflection of Light*. DOI: 10.5281/zenodo.22010983.